Unit 5: Gravitation

Complete notes for Unit 5 of *A Textbook of Physics, Grade IX* (Khyber Pakhtunkhwa Textbook Board, Peshawar) - every definition, formula and worked example from the book, plus the full exercise solved: MCQs with an answer key, all seven numericals step by step, and short answers to the conceptual questions.

> **How to use these notes.** Read one section at a time, copy each formula into your notebook, and redo every solved example yourself before looking at the answer. The last pages are pure exam practice - the book's own exercise, solved.

What this unit covers

Book section	Topic	Book pages
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Exercise	MCQs, conceptual, comprehensive and numerical questions	145-147

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After studying this unit, you should be able to

- State Newton's law of gravitation.
- Explain that gravitational forces are consistent with Newton's third law.
- Explain gravitational field as an example of a field of force.
- Define weight as the force on an object due to a gravitational field.
- Calculate the mass of Earth by using the law of gravitation.
- Solve problems using Newton's law of gravitation.
- Explain that the value of g decreases with altitude from the surface of Earth.
- Discuss the importance of Newton's law of gravitation in understanding the motion of satellites.

These are the student learning outcomes printed on the unit opener (book page 129). The Earth and the Moon orbit around each other and are kept together by their gravitational interaction - that one idea runs through the whole unit.

Why gravity matters at every scale

The study of gravity runs from Galileo's experiments on free fall in the seventeenth century to Stephen Hawking's work on black holes. The force that binds together structures from the Moon all the way to galaxy clusters is the gravitational force.

Gravity is an **action-at-a-distance force**: it acts even when the interacting objects are not in physical contact. It holds us on Earth, and it also reaches out across intergalactic space and acts between galaxies. Our Milky Way is a disk-shaped collection of gas, dust and billions of stars; Earth sits near the edge of the disk, about 26,000 light-years (2.5×10^{20} m) from its central bulge.

5.1 Law of universal gravitation

> **Newton's law of universal gravitation:** Every body in the universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between their centres.

For two spherical bodies of masses m_1 and m_2 separated by distance r between their centres:

$$F_g = G\, m_1\, m_2 / r^2$$

Equation 5.1 - G is the gravitational constant

- F_g proportional to m_1 x m_2 - bigger masses attract more strongly.
- F_g proportional to $1 / r^2$ - this is an **inverse-square** law.
- $G = 6.67 \times 10^{-11}$ Nm^2kg^{-2}. Cavendish measured the tiny attraction between suspended spheres and large spheres on a stand with a **torsion balance**, observing the twist in the fibre.
- G does **not** depend on the medium (vacuum, air, water or concrete) between the two bodies.

Gravitation and Newton's third law

The force on mass m_2 due to m_1 is F_{12}, and the force on m_1 due to m_2 is F_{21}. The two forces are equal in magnitude and opposite in direction, so $F_{12} = -F_{21}$: gravitational forces form an **action and reaction pair**.

Earth pulls the Moon, and the Moon pulls Earth with a force of equal magnitude. At Earth's surface, Earth pulls down on a 1.0 kg mass with a force of 9.8 N - and the 1.0 kg mass pulls upward on Earth with a force of 9.8 N.

The inverse-square rule in action

Because the force varies as $1 / r^2$ (the book's Graph 5.1):

Distance becomes	Force becomes
2 x r	F / 4
3 x r	F / 9
r / 2	4 x F

EXAMPLE 5.1 - CAN YOU ATTRACT ANOTHER PERSON GRAVITATIONALLY?

Given: $m_1 = 40$ kg, $m_2 = 45$ kg, $r = 0.5$ m, $G = 6.67 \times 10^{-11}$ Nm^2kg^{-2}. **Find:** F_g.

$F_g = 6.67 \times 10^{-11} \times (40 \times 45) / (0.5)^2 = 6.67 \times 10^{-11} \times 1800 / 0.25$

$F_g = 4.8 \times 10^{-7}$ N - we do attract each other, but a force of about 10^{-7} N is unnoticeably small unless extremely sensitive instruments are used. Even ships and buildings have very small gravitational attraction.

ASSIGNMENT 5.1 SOLVED - GRAVITATIONAL FORCE ON THE MOON

Mass of Earth = 6×10^{24} kg, mass of Moon = 7.4×10^{22} kg, distance = 3.84×10^5 km = 3.84×10^8 m.

$F = 6.67 \times 10^{-11} \times (6 \times 10^{24} \times 7.4 \times 10^{22}) / (3.84 \times 10^8)^2$

$F = 2.0 \times 10^{20}$ N - the force Earth exerts on the Moon (and the Moon exerts on Earth).

Watch the units. Distances given in kilometres must be changed to metres before putting them into the formula, and the distance is always measured centre to centre.

5.2 Mass of Earth

Knowing G let scientists calculate the mass of Earth, the Sun and other celestial objects. Place an object of mass m_o on Earth's surface. Its distance from Earth's centre is Earth's radius r_E, and Earth (mass m_E) attracts it with:

$$F_g = G\, m_o\, m_E / r_E^2$$

This force of gravity is the object's weight: $F_g = W = m_o\, g$. Comparing the two:

$$m_o\, g = G\, m_o\, m_E / r_E^2 \quad \text{so} \quad g = G\, m_E / r_E^2$$

$$\mathbf{m_E = g\, r_E^2 / G}$$

Equation 5.3 - mass of Earth (or any planet)

With $g = 9.8\ ms^{-2}$, $r_E = 6.4 \times 10^6$ m and $G = 6.67 \times 10^{-11}\ Nm^2kg^{-2}$:

$m_E = 9.8 \times (6.4 \times 10^6)^2 / (6.67 \times 10^{-11}) = 9.8 \times 4.1 \times 10^{13} / 6.67 \times 10^{-11}$

$m_E = 6 \times 10^{24}$ kg. The same method works for any planet or star whose g and radius are known.

EXAMPLE 5.2 - MASS OF JUPITER

$r_J = 7.15 \times 10^7$ m, $g_J = 24.7$ N/kg. $m_J = g_J\, r_J^2 / G = 24.7 \times (7.15 \times 10^7)^2 / 6.673 \times 10^{-11}$

$m_J = 1.89 \times 10^{27}$ kg - about 300 times the mass of Earth.

ASSIGNMENT 5.2 SOLVED - MASS OF THE MOON

$r = 1.74 \times 10^6$ m, $g = 1.6\ ms^{-2}$. $m = 1.6 \times (1.74 \times 10^6)^2 / 6.67 \times 10^{-11}$

$m = 7.3 \times 10^{22}$ kg

Solar system data (book Table 5.1)

Planet	Radius (km)	Mass (10^{24} kg)	g (m/s^2)	Orbital period (yr)
Mercury	2,400	0.330	3.7	0.241
Venus	6,050	4.87	8.9	0.615
Earth	6,370	5.97	9.8	1
Mars	3,400	0.642	3.7	1.88
Jupiter	71,500	1,890	24.7	11.9
Saturn	60,300	568	9.0	29.4
Uranus	25,600	86.8	8.7	83.8
Neptune	24,800	102	11.0	164
Sun	696,000	1,990,000	274	--

> **Values to remember:** G = 6.67 x 10^{-11} Nm^2kg^{-2}; radius of Earth r_E = 6.4 x 10^6 m; mass of Earth = 6 x 10^{24} kg; g on Earth = 9.8 ms^{-2}.

5.3 Gravitational field

Gravity is a field force. Newton could describe gravity mathematically but could not explain how objects exert forces without contact. Field theory answers this: masses create a **gravitational field** in the space around them (just as charges create an electric field and magnets a magnetic field), and a gravitational force is an interaction between a mass and the field created by other masses.

> **Gravitational field strength g:** the gravitational force exerted on a unit mass at that point, g = F_g / m. It is a vector, pointing in the direction of the gravitational force - towards the centre of the attracting mass.

For an object free to accelerate under gravity alone, Newton's second law gives a = F / m, so the field strength at a point equals the **free-fall acceleration** there: g = 9.8 m/s^2 on Earth's surface. They are equivalent but not the same thing: hanging an object from a spring scale measures field strength, because the resting mass shows no measurable acceleration.

Weight changes with location

> **Weight:** the force on an object due to a gravitational field - mass times gravitational field strength, W = m g. Mass (in kg) never changes with location; weight (in newtons) does.

From $g = G m_E / r_E^2$ (Equation 5.4), using $m_E = 6 \times 10^{24}$ kg and $r_E = 6.4 \times 10^6$ m gives $g = 9.77$ ms^{-2}, about 9.8 ms^{-2}. Notice that **g does not depend on the mass of the falling body** - light and heavy bodies fall towards Earth with the same acceleration. g depends only on the planet's mass and radius.

Earth is not a perfect sphere, and mountains and trenches change our distance from its centre, so g - and therefore our weight - varies slightly from place to place. On any planet, g and weight depend on that planet's mass and radius. Geophysicists use sensitive **gravimeters** to detect small variations in g when searching for minerals and oil: gold and silver deposits increase g, oil and gas deposits decrease it.

EXAMPLE 5.3 - VALUE OF G AT THE SUN'S SURFACE

$m_S = 1.99 \times 10^{30}$ kg, $r_S = 6.96 \times 10^8$ m. $g = G m_S / r_S^2 = 6.67 \times 10^{-11} \times 1.99 \times 10^{30} / (6.96 \times 10^8)^2$

g = 274 ms^{-2} - about 30 times Earth's g, although the Sun's mass is almost a million times Earth's.

ASSIGNMENT 5.3 SOLVED - FREE FALL AT THE MOON

Moon: $m = 7.35 \times 10^{22}$ kg, $r = 1.74 \times 10^6$ m. $g = 6.67 \times 10^{-11} \times 7.35 \times 10^{22} / (1.74 \times 10^6)^2 =$ **1.62 ms^{-2}**. Astronaut (65.0 kg, 637 N on Earth): weight on Moon = 65 x 1.62 = **105 N** - about one-sixth of his Earth weight, while his mass stays 65.0 kg.

5.4 Variation of g with altitude

The value of g at a place depends on its distance from Earth's centre: the greater the distance, the smaller g. At a height h above the surface, the distance from the centre becomes (r_E + h), so:

$$g_h = G\, m_E / (r_E + h)^2$$

Since at the surface $G\, m_E = g\, r_E^2$, substituting gives:

$$\mathbf{g_h = g\, r_E^2 / (r_E + h)^2}$$

Equation 5.5 - g falls as altitude rises

As we move away from Earth's centre, g decreases. The change is significant only at very large distances:

Altitude (km)	g (m/s^2)	Example
0.0	9.8	Mean Earth radius
8.8	9.8	Mt. Everest
36.6	9.7	Highest manned balloon
400	8.7	Space shuttle orbit
35,700	0.2	Communications satellite

EXAMPLE 5.4 - METEOROID IN EARTH'S FIELD

h = 4.4 x 10^7 m = 44 x 10^6 m, r_E = 6.4 x 10^6 m. g_h = 9.8 x (6.4 x 10^6)^2 / (50.4 x 10^6)^2

g_h = 0.16 ms^{-2}

ASSIGNMENT 5.4 SOLVED - G AT TWO HEIGHTS

At 1,000 km: g_h = 9.8 x (6,400 / 7,400)^2 = **7.33 ms^{-2}**. At 35,900 km: g_h = 9.8 x (6,400 / 42,300)^2 = **0.22 ms^{-2}**.

Point to ponder (book): g on the Moon is about 1/6 of Earth's g - does that mean the Moon's mass is 1/6 of Earth's? No: g also depends on radius squared, and the Moon's radius is much smaller, so its mass works out to about 1/81 of Earth's.

5.5 Motion of artificial satellites

> **Satellite:** a naturally occurring or man-made object that orbits another, larger body by the force of gravity. Moons are natural satellites. An object purposely placed into orbit around Earth, another planet or a star is an **artificial satellite**.

The first artificial satellite was launched in 1957; since then thousands have been sent up for communications (internet, mobile phones, TV), weather study, military purposes and scientific research.

What keeps a satellite up?

Its high speed. A satellite is carried to a high altitude and then accelerated by rockets to a sufficiently high **tangential speed**. If the speed is too low, it falls back to Earth; if it is too high, it moves in an elliptical orbit or escapes Earth's gravity, never to return. Satellites are usually put into circular (or nearly circular) orbits because these need the least take-off speed. If a satellite in orbit stopped moving, it would fall directly to Earth.

Orbital velocity

For a satellite of mass m_s in a circular orbit of radius $r = r_E + h$ with velocity v, gravity provides the centripetal force:

$$F_g = G\, m_E\, m_s / r^2 \quad \text{and} \quad F_c = m_s\, v^2 / r \quad \text{so} \quad m_s\, v^2 / r = G\, m_E\, m_s / r^2$$

$$\mathbf{v = \text{sqrt}(\, G\, m_E / (r_E + h)\,)}$$

Equation 5.6 - orbital speed; m_s cancels out

Orbital speed depends on the mass of Earth and on the distance from Earth's centre - **not** on the mass of the satellite. At a given height, large and small satellites must have the same orbital speed.

Geostationary satellites sit in a circular orbit 3.59×10^7 m = 35.9×10^6 m above the surface. Equation 5.6 gives v = sqrt($6.673 \times 10^{-11} \times 6 \times 10^{24} / (35.9 + 6.4) \times 10^6$) = **$3.07 \times 10^3$ ms^{-1} = 3.07 km/s**.

EXAMPLE 5.5 - SPEED OF EARTH AROUND THE SUN

Mass of Sun $m_S = 1.99 \times 10^{30}$ kg, radius of Earth's orbit $r = 1.5 \times 10^{11}$ m.

$v = \sqrt{G m_S / r} = \sqrt{6.673 \times 10^{-11} \times 1.99 \times 10^{30} / 1.5 \times 10^{11}}$

$v = 3.0 \times 10^4$ m/s = 30 km/s - the same equation measures the velocity of any planet or celestial object in a circular path.

ASSIGNMENT 5.5 SOLVED - ORBITAL SPEED OF A SATELLITE

Height 2,000 km, so $r = 6{,}400 + 2{,}000 = 8{,}400$ km $= 8.4 \times 10^6$ m.

$v = \sqrt{6.67 \times 10^{-11} \times 6 \times 10^{24} / 8.4 \times 10^6} = \sqrt{4.76 \times 10^7}$

$v = 6.9 \times 10^3$ m/s = 6.9 km/s

Formula sheet and key points

Quantity	Formula / value
Gravitational force	$F_g = G m_1 m_2 / r^2$
Gravitational constant	$G = 6.67 \times 10^{-11}$ Nm^2kg^{-2}
Mass of a planet	$m = g r^2 / G$
Field strength on a planet	$g = G m / r^2$
Weight	$W = m g$
g at height h	$g_h = g r_E^2 / (r_E + h)^2$
Orbital velocity	$v = \sqrt{G m_E / (r_E + h)}$
Mass / radius of Earth	6×10^{24} kg / 6.4×10^6 m

- **Book key points:** g is the acceleration of bodies falling freely towards Earth, and it decreases as altitude increases. Artificial satellites revolve round Earth under gravity, which provides the necessary centripetal force.

Exam traps: (1) Mass stays the same everywhere - only weight changes. (2) In orbital speed, r is measured from Earth's centre: always add h to r_E. (3) Doubling both masses makes the force 4 times, not 2 times. (4) g at the equator is smaller than at the poles.

Exercise MCQs - with answer key (book page 145)

1. Two masses are separated by distance r. If both masses are doubled, the force of interaction changes by a factor of - **B. 4** (force is proportional to the product of the masses: 2 x 2 = 4).
2. The radius of Earth r_E is - **D. 6.4×10^6 m**.
3. The SI unit for gravitational constant G is - **B. Nm^2kg^{-2}**.
4. What is the mass on planet Mercury of an object that weighs 784 N on Earth's surface? - **A. 80.0 kg** (m = W / g = 784 / 9.8 = 80 kg; mass is the same on Mercury).
5. An object orbits a planet with orbital speed v. If the radius is the same and the planet's mass is increased 4 times, the orbital speed changes by a factor of - **A. 2** (v is proportional to the square root of mass: sqrt(4) = 2).
6. The value of g at the surface of the Moon is - **B. 1.63 ms^{-2}**.
7. The value of g _____ with altitude - **B. decreases**.
8. When a body is moved from sea level to the top of a mountain, there is a change in the body's - **B. weight** (mass stays the same; g, and so weight, falls slightly).
9. The value of g at the equator is - **C. smaller than at the poles** (Earth's radius is greatest at the equator).

Answer key: 1-B 2-D 3-B 4-A 5-A 6-B 7-B 8-B 9-C

Solved numerical questions (book page 147)

1. Pluto and Charon. $m_1 = 1.25 \times 10^{22}$ kg, $m_2 = 1.9 \times 10^{21}$ kg, $r = 1.96 \times 10^4$ km $= 1.96 \times 10^7$ m. $F = 6.67 \times 10^{-11} \times (1.25 \times 10^{22} \times 1.9 \times 10^{21}) / (1.96 \times 10^7)^2$ **$= 4.1 \times 10^{18}$ N**.

2. Field strength on Mars. $m = 6.4 \times 10^{23}$ kg, $r = 3.4 \times 10^6$ m. $g = 6.67 \times 10^{-11} \times 6.4 \times 10^{23} / (3.4 \times 10^6)^2$ **$= 3.7$ ms^{-2}**.

3. Acceleration due to gravity on Titan. $m = 1.35 \times 10^{23}$ kg, $r = 2{,}570$ km $= 2.57 \times 10^6$ m. $g = 6.67 \times 10^{-11} \times 1.35 \times 10^{23} / (2.57 \times 10^6)^2$ **$= 1.36$ ms^{-2}**.

4. Altitude where g = 4.9 ms^{-2}. $g_h / g = 1/2$, so $(r_E / (r_E + h))^2 = 1/2$, giving $r_E + h = 1.414\, r_E$ and $h = 0.414 \times 6.4 \times 10^6$ m **$= 2.65 \times 10^6$ m (about 2,650 km)**.

5. Satellite 225 km above Earth. $r = 6.4 \times 10^6 + 0.225 \times 10^6 = 6.625 \times 10^6$ m. $v = \sqrt{6.67 \times 10^{-11} \times 6 \times 10^{24} / 6.625 \times 10^6}$ **$= 7.77 \times 10^3$ m/s (7.8 km/s)**.

6. Orbital speed of the Moon. $r = 3.8 \times 10^8$ m, $m_E = 6 \times 10^{24}$ kg. $v = \sqrt{6.67 \times 10^{-11} \times 6 \times 10^{24} / 3.8 \times 10^8}$ **$= 1.03 \times 10^3$ m/s (1.03 km/s)**.

7. Altitude of the Hubble telescope. $v = 7.6 \times 10^3$ m/s. From $v^2 = G\, m_E / r$, $r = 6.67 \times 10^{-11} \times 6 \times 10^{24} / (7.6 \times 10^3)^2 = 6.93 \times 10^6$ m. Altitude $h = r - r_E = 6.93 \times 10^6 - 6.4 \times 10^6$ **$= 5.3 \times 10^5$ m (about 529 km)**.

Conceptual questions - short answers (book page 146)

1. **Why do we not feel ourselves gravitating toward nearby massive buildings?** The force exists, but G is extremely small, so the attraction between a person and even a large building is far too tiny to feel (compare Example 5.1: about 10^{-7} N between two people).
2. **Does the Sun exert a larger force on Earth than Earth exerts on the Sun?** No. By Newton's third law the two forces are equal in magnitude and opposite in direction; the Sun barely moves because its mass - and so its inertia - is so much greater.
3. **Importance of G, and why it is difficult to calculate?** G turns the proportionality into an equation, lets us find forces between any masses and even weigh planets. It is difficult because the forces between laboratory masses are extremely small and need a very sensitive torsion balance.
4. **If Earth expanded to a larger radius with no change in mass, how would your weight be affected? If it shrank?** g is proportional to 1 / r^2: expanding would decrease your weight; shrinking would increase it.
5. **If Earth's mass doubled but its radius stayed the same?** g is proportional to mass, so g - and your weight - would double. Your mass would not change.
6. **Why do lighter and heavier objects fall at the same rate?** $g = G\, m_E / r_E^2$ does not contain the falling object's mass, so all bodies get the same acceleration (ignoring air resistance).
7. **g changes with location, so why use 9.8 ms^{-2} everywhere?** The variation across Earth's surface is very small, so one average value is accurate enough for ordinary calculations.
8. **The Moon is attracted by Earth - why does it not fall onto Earth?** The Moon has a large tangential velocity. It is continually falling towards Earth, but its sideways motion keeps it in orbit, just like an artificial satellite.
9. **Why must larger and smaller satellites at the same height have the same orbital speeds?** In $v = \sqrt{G\, m_E / (r_E + h)}$ the satellite's mass cancels, so speed depends only on height.

Comprehensive questions - what each answer must contain

1. State and explain the law of universal gravitation, and show it obeys Newton's third law - statement, equation 5.1, G, and the equal-and-opposite pair $F_{12} = -F_{21}$.
2. Determine the mass of Earth - the derivation of $m_E = g\, r_E^2 / G$ and the value 6×10^{24} kg.
3. Gravitational field and field strength; show weight changes with location - field idea, $g = F / m$, $W = m\, g$ with g depending on planet and place.
4. How g changes with higher altitude - derivation and equation 5.5.
5. Derive the orbital speed formula - set gravity equal to centripetal force and reach equation 5.6.

End of Unit 5 notes. Notes prepared by **Shah Fahad** - 03409090945